

Task: DZI - Plots

Lab 6. Available memory: 512 MB.

Date: 07.12.2025

Description

Bajtazar plans to buy a plot of land. He has received many location proposals, but his primary selection criterion will be the number of trees on the plot.

Help Bajtazar write a program that, given the coordinates of trees in Bajtocja $P = [(x_1, y_1), \dots, (x_n, y_n)]$ and the coordinates of plots, calculates the number of trees contained in each plot.

For a query about a plot (x_l, x_r, y_b, y_t) , you must calculate:

$$|\{i \in \{1, \dots, n\} : x_l \leq x_i \leq x_r \text{ and } y_b \leq y_i \leq y_t\}|$$

Input

The first line contains two integers n and m ($1 \leq n \leq 400\,000$, $1 \leq m \leq 200\,000$), denoting the number of trees and plots. The next n lines provide the coordinates of the trees; each line contains two integers x_i and y_i ($0 \leq x_i \leq 10^9$, $0 \leq y_i \leq 10^9$). More than one tree may exist at a single point. The next m lines provide the coordinates of the plots; each line contains four integers x_l, x_r, y_b, y_t ($0 \leq x_l \leq x_r \leq 10^9$, $0 \leq y_b \leq y_t \leq 10^9$).

Output

The program should print m lines. In the i -th line, write the number of trees contained in the i -th plot.

Example

Input:

```
5 3
1 1
1 6
2 2
3 3
4 2
0 10 0 10
2 5 3 4
3 4 2 5
```

Output:

```
5
1
2
```